

Fishing for Sustainability

The Ultimate Goal

Maximize your fishing yields by the end of the game.

Setup

Groups

- Form groups of 4-6 students each for the game.
- Each group will receive:
 - 1 bowl (your *reservoir*)
 - 30 fish placed in that reservoir
- Each person will receive:
 - 1 cup (your *boat*)

Important Biological Rule

- After each fishing round in this game, any fish left in the reservoir double in number, but only up to a maximum of 30 fish (*carrying capacity*).

Round 1: Open Access Fishing

Scenario

- Each of you represents a fisher in a shared fishery aiming to catch as many fish as possible.
- You will have no means of communication or coordination with other fishers in your fishery.
- After this fishing season (the end of the round), the fish will enter a reproductive phase, and any that are not harvested will produce one offspring each.

Instructions

1. Privately record how many fish you will catch this round (up to 6 allowed).
- ***Remember, no communicating ahead of time during this round!***
2. Simultaneously reveal everyone's card with their number.
 3. Distribute each person's catch into their boat.
- ***If the cumulative group catch exceeds the number of fish available, distribute the fish until the reservoir is empty (first-come, first-served)***

4. Give the fish a reproductive season.
 - a. Count the remaining fish.
 - b. Add new fish so that the remaining number has doubled in size.
 - c. Record these data (*fish caught*, *fish remaining*)

Pause and record a prediction: *How long could the fishery sustain with this type of open-access fishing?* Write your response (1-2 sentences) on the back of the notecard with your name and today's date.

Context and Theory

Did you notice how quickly the shared resource (your fishery) can become depleted when everyone acts in their own self-interest? This situation is an example of a social dilemma—a scenario where individuals benefit most by acting selfishly, but if everyone does this, the group as a whole suffers (*The Prisoner's Dilemma*).

This dynamic is often called the Tragedy of the Commons—a term used in conservation biology to describe what happens when shared resources (like fisheries, forests, or wildlife populations) are overused and degraded because no one has a personal incentive to conserve them.

Before continuing to the next round, watch this short video explaining the Tragedy of the Commons.

Modifying our Approach

Now we will adjust our rules so that you can act more strategically to balance your short-term choices with the population's sustainability in mind. Pause for a moment and try to predict the **maximum number of fish** each member of your group could afford to harvest in one season without reducing the overall population in the long term. This concept is known as the *Maximum Sustainable Yield (MSY)*, and it relies on the assumption that your fish will maintain a consistent reproductive rate from one season to the next. In our case, the population doubles each round, and so the MSY is reached when enough fish are left behind to fully replenish the stock.

>Pause to watch this youtube video before proceeding.

In the next round we will use a strategy called **Fixed Quota Harvesting** to improve the sustainability of our yield. This is a management strategy where the **total amount harvested is limited** to a set number (e.g., 15 fish per round). The quota must be shared among users, and exceeding it can lead to collapse or penalties.

Round 2: Fixed-Quota Harvesting

Scenario

- You may now work together to set a **total fixed quota** (i.e., one total for the group) to sustainably harvest each round (recommended quota around the MSY, e.g., 15 fish).
- You still may not discuss how many fish each individual plans to catch to contribute to this quota, and your goal is still to maximize your long-term yields as an individual.

Instructions

1. Discuss openly for 1–2 minutes and agree on a fixed quota (e.g., “Our total catch limit is exactly 15 fish per round.”).
2. Privately record how many fish you will catch this round (up to 6 allowed).

- *Remember, still no communicating this part*

3. Simultaneously reveal everyone’s card with their number.
4. This time, if the **total catch exceeds the quota**, the fishery immediately “collapses”, and no fish can reproduce.

- If your group’s total catch is equal or less than quota, you distribute each person’s catch again, and reproduction proceeds normally (fish left double, up to 30).
- Record these results (*fish caught, fish remaining*)

5. Pause and record a reflection: *Was it challenging or easy to respect quotas?*

Context and Theory

How else could you manage your group’s collective harvesting to maximize your long-term yields? One option is known as ***Fixed Effort Harvesting***. In this approach, the game manager specifies the level of effort (for example the number of fishers/hunters or length of a fishing/hunting season), rather than the number of animals to be harvested. This strategy relies on the assumption that harvest rate increases with the amount of effort, so for our purposes, we can simply target the quota of fish per person.

Round 3: Fixed-Effort Harvesting

Scenario

- You may now work together to set a **fixed effort quota** (i.e., a per-person fishing limit) to sustainably harvest each round (recommended quota around the MSY, e.g., 2 fish).

Instructions

1. No need for any secrecy this time. Instead, you need to collaborate as a group to collectively decide on how many fish to distribute to each person and then pass them out.
2. Give the remaining fish a reproductive season.

- a. Count the remaining fish.
- b. Add new fish so that the remaining number has doubled in size.
- c. Record these data (*fish caught, fish remaining*)

3. Pause and record a reflection: *Did fixed-effort make sustainability easier or harder to maintain?*